

Unequal Job Opportunities and the Measurement of Welfare Inequality

Hisham Haydar

LISER and University of Luxembourg

France 2016, EU-SILC priced through EUROMOD.

Thank you. This paper asks how much of the inequality in money-metric well-being among single adults in France is attributable to unequal job opportunities, and what happens to that attribution when a model omits opportunities altogether.

Two questions.

(1) How much money-metric well-being inequality is attributable to unequal job opportunities rather than to preferences?

(2) When a model omits heterogeneous opportunities, how much of that inequality is re-attributed to preferences?

Consider two people in the same job, at the same hours and the same wage. One chose it from a wide set of alternatives; the other had almost nothing else available. A model in which every household faces the same choice set cannot distinguish them: whatever the choice set cannot express is absorbed by the taste parameters. The first question is quantitative. The second is about what a conventional model gets wrong.

Three literatures, one gap.

Money-metric welfare with heterogeneous preferences

Fleurbaey and Maniquet (2005, 2011); Decoster and Haan (2015); Bargain, Decoster, Dolls, Neumann, Peichl and Siegloch (2013); Jacquet, Jia and Thoresen.

Random-utility random-opportunity labour supply

Dagsvik (1994); Aaberge, Colombino and Strøm (1999); Aaberge and Colombino (2013); Dagsvik and Jia (2016); Capéau et al.

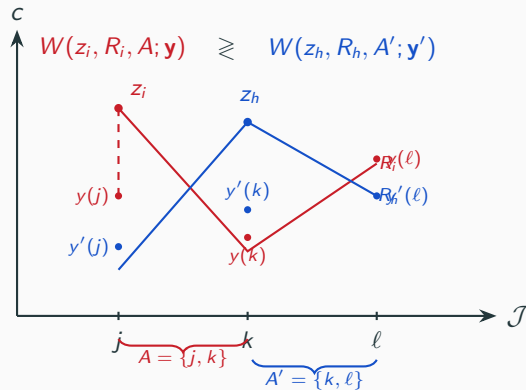
Inequality of opportunity

Roemer (1998); Bourguignon, Ferreira and Menéndez (2007); Ferreira and Gignoux (2011).

This paper: household-specific latent job-opportunity distributions carried into a preference-respecting money-metric inequality decomposition, in both counterfactual directions.

Three literatures meet here. Money-metric welfare analysis with heterogeneous preferences — Decoster and Haan is the closest empirical precedent — represents individual constraints by wages, non-labour income and the tax-benefit budget, without a household-specific distribution of available jobs. The RURO labour-supply tradition estimates exactly such a distribution, but has used it for behaviour and reform simulation, not for inequality decomposition. And the inequality-of-opportunity literature decomposes outcomes into circumstances and effort without a structural model of the job market. The gap is the opportunity object: I estimate it, and carry it into the welfare decomposition.

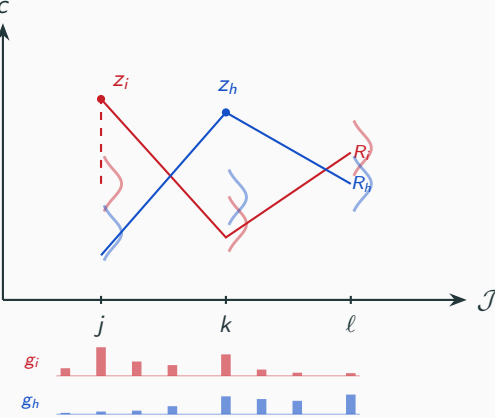
A job is a package; welfare is compared on the set of reachable jobs.



W^1 : the uniform pay across the jobs you can reach that would leave you as well off - Haydar and Maniquet (companion paper).

The companion theory paper, with Francois Maniquet, characterizes a family of welfare measures for economies where jobs are discrete packages and each person can reach only a subset of them. The measure I take to the data, W^1 , asks: at what uniform pay across the jobs you can reach would you be exactly as well off as you are now? It neutralizes differences in pay across the jobs in your set, so that pay luck does not enter; differences in the set itself remain in the measure - and they are exactly what the decomposition then attributes.

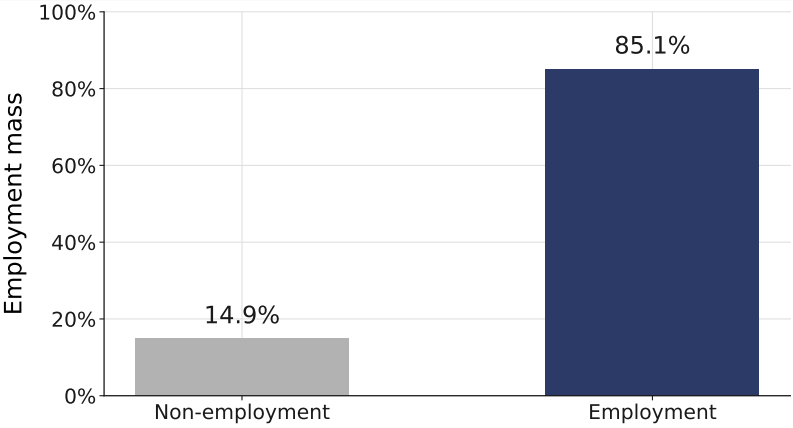
Empirically the set is an estimated distribution and the pay is an offer density.



estimated jointly with preferences; nodes are integration support, not a list of jobs.

In the data no one observes the set of jobs a household could have had. What can be estimated is a density over latent job packages — hours, occupation and wage — that says how available each package is to a given household. The picture changes in exactly two places: the set becomes a distribution, the pay becomes a density. Everything downstream is a functional of that estimated distribution.

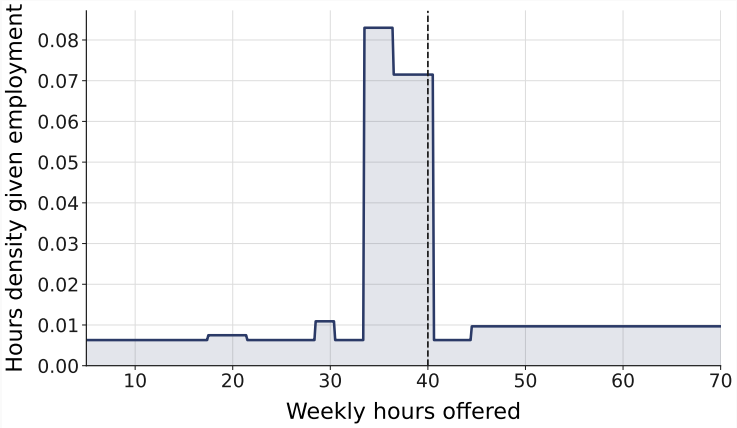
For one household: its estimated employment mass, hours weights, occupation shares and wage-offer density



estimated opportunity distribution; nodes are integration support, not a list of jobs

Here the schematic becomes an estimate. For this household the model gives an 85 per cent chance that an offer is taken up, hours weights concentrated on the 35-hour and full-time bands, sex-specific occupation shares, and a log-normal wage-offer density centred at eleven euros. The welfare measure is a functional of exactly these four objects.

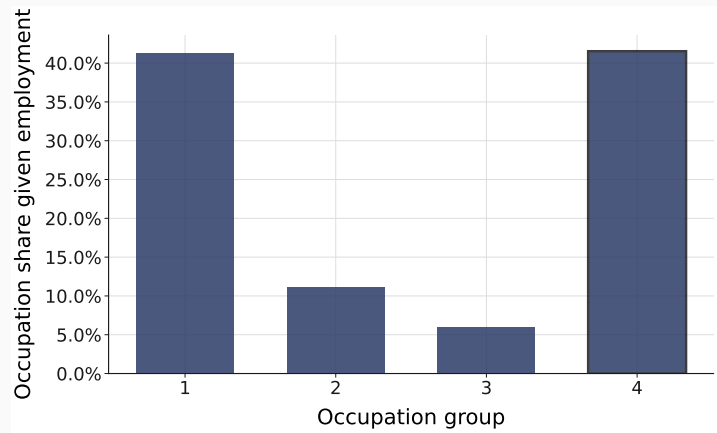
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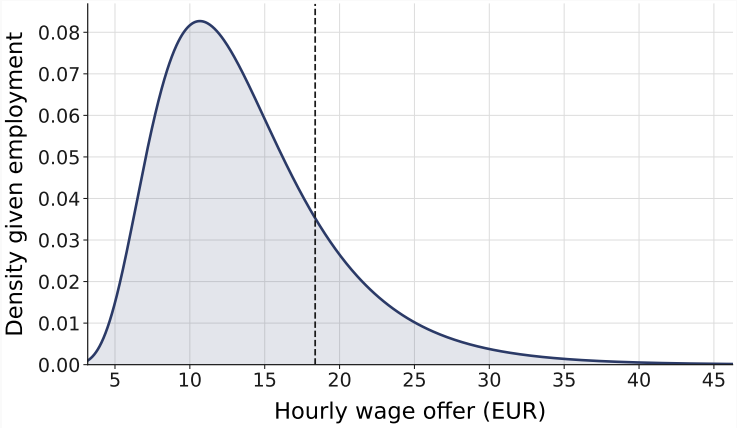
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Preferences and an opportunity density, in one likelihood.

$$u_{ij} = \beta_{\ell}(x_i) \text{BC}(l_{ij}; \theta_{\ell}) + \text{BC}(c_{ij}; \theta_c)$$

$$g_{ij} = \underbrace{g^E}_{\text{whether work is available}} \cdot \underbrace{g^H}_{\text{at which hours}} \cdot \underbrace{g^{\text{Occ}}}_{\text{in which occupation}} \cdot \underbrace{g^W}_{\text{at what pay}}$$

41 estimated structural parameters.

Box–Cox preferences over consumption and leisure, with age and children shifting the leisure weight. The opportunity density factorizes into an access margin, a piecewise-constant hours factor with a peak at the statutory 35-hour week, sex-specific occupation availability, and an occupation-conditioned log-normal wage-offer location with a common education and experience gradient. The two objects are estimated together, which is what prevents a taste parameter from absorbing an availability parameter.

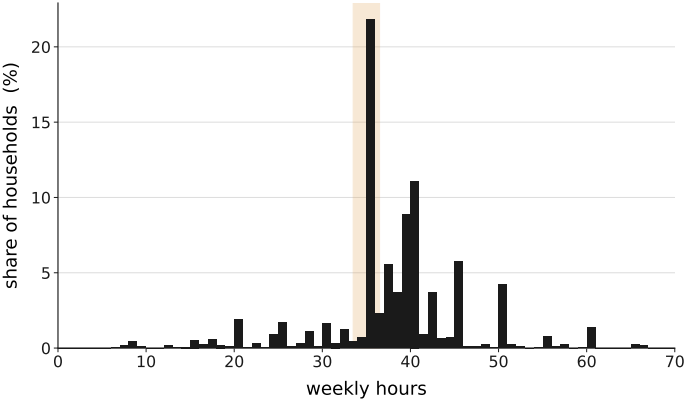
Each household: its observed job plus 100 drawn alternatives, all priced through the tax-benefit system.

$$V_{ij} = \underbrace{u_{ij}}_{\text{preferences}} + \underbrace{\log g_{ij}}_{\text{availability}} - \underbrace{\log q_{ij}}_{\text{sampling correction}}$$

EUROMOD FR 2016 → disposable income for every one of 157,055 alternatives.

The choice set is not enumerable, so it is sampled: 100 latent jobs per household from a known proposal, plus the observed job inserted deterministically. The sampling correction makes the sampled-set likelihood consistent for the full model; the proposal is computation, the opportunity density is the economics. Every alternative is priced through EUROMOD, so consumption is computed, not surveyed, and the tax-benefit system enters the opportunity set directly.

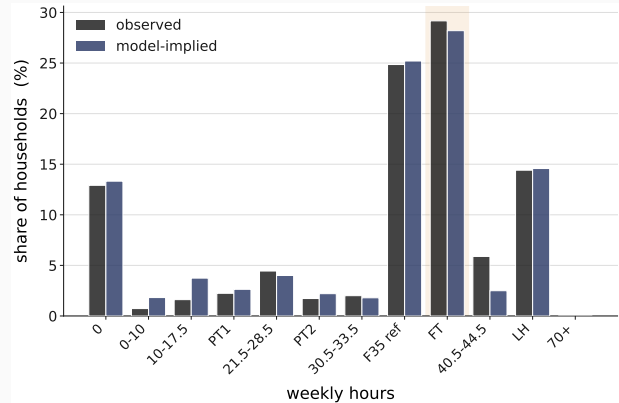
France 2016: 1,555 single-adult households; one in four employed at exactly 35 hours.



The estimation sample is single-adult households aged 20 to 60. The institution that dominates the hours distribution is visible: a quarter of households are employed in the 35-hour band. A model without an explicit peak there misses the hours distribution badly; with it, the fit is close.

EU-SILC 2016; weights; 1,348 employed.

The model reproduces the hours distribution, employment and occupation shares.



35-hour band 24.9% observed, 25.2% model-implied; employment 87.1% / 86.7%; mean absolute band error 0.008.

Observed against model-implied shares by hours band: the statutory band is matched to within half a point, employment and the four occupation groups to within a point. The one visible misfit is the bottom of the offered-wage distribution, which the model over-predicts by about five points; I tested an hours-conditioned repair and it did not improve the fit.

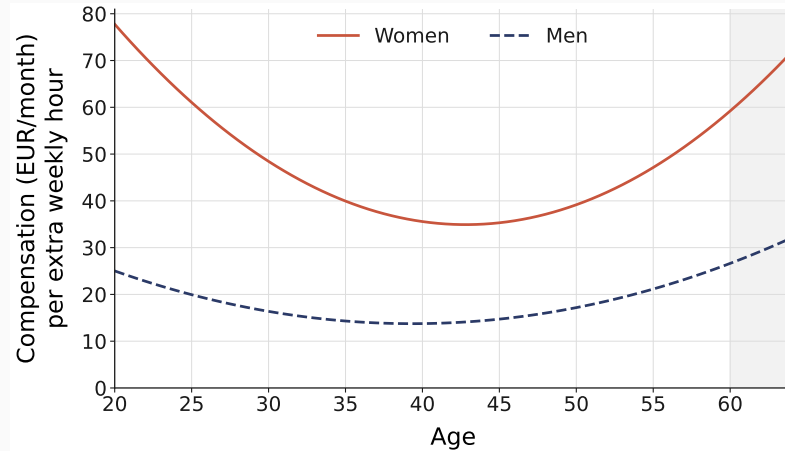
Estimated indifference curves and the price of an hour, by age and sex



Box-Cox preferences at the estimated parameters; own budget; one household per sex

These are the estimated preferences themselves. The curves are steep at long hours and flat at short hours, and the leisure weight is U-shaped in age with its minimum around forty. The price of one more weekly hour - the consumption that keeps a household indifferent - is what the welfare measure trades against opportunities.

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What the data pin down, and what remains conditional.

pinned

the preference block;

the four availability factors;

the 35-hour peak.

conditional

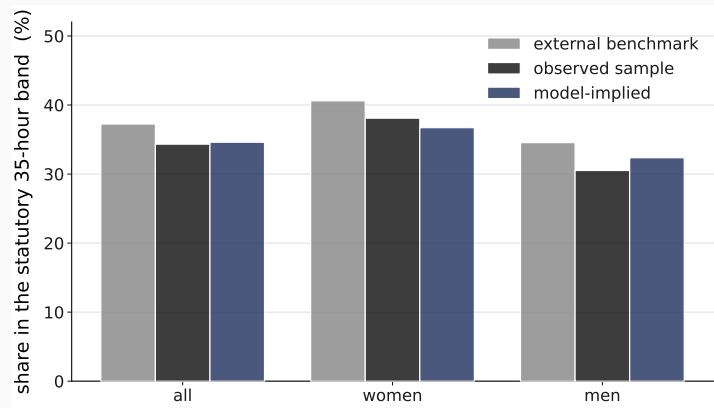
access versus capability (not separately identified);

persistent unobserved heterogeneity (three pre-specified extensions not recoverable from one cross-section).

structural and model-conditional; not causal.

Two things stay conditional and I say so. The access density combines personal capability and market availability; the data do not separate them. And persistent household heterogeneity — a random leisure intercept, a persistent access frailty, and dependence between the wage residual and preferences — could not be recovered in synthetic experiments, each for a measured reason: the objects they load on vary too little within a household's own choice set.

An independent source puts the 35-hour concentration at 37%; the sample at 34%; the model at 35%.



The Labour Force Survey never enters the estimation. Its share of employment in the 35-hour band, among the bands the model prices, is 37 per cent; the estimation sample gives 34 and the model 35, and the survey's ordering of women above men is reproduced by a model with no sex-specific hours coordinate. The survey also documents underemployment declining in hours and a large positive gap between desired and actual hours at short hours — context for the thin short-hours opportunities the model estimates, not a moment it matches.

Labour Force Survey 2016; validation only; no external moment enters the likelihood.

Equivalent income at a common reference pay, under four preference–environment states.

$$W_i^1 = w : \quad \Omega_i(\text{flat pay } w \text{ on own set}) = \Omega_i(\text{actual})$$

i.e. the uniform pay w over the household’s own opportunity distribution that reproduces its attained expected welfare.

	own environment	reference environment
own preferences	I_{00}	I_{01}
reference preferences	I_{10}	I_{11}

reference environment = common job access, earning opportunities and budget; the same coalition’s set and preferences on both sides of the inversion; common quadrature support.

Welfare is the money-metric level: the uniform pay across the jobs a household can reach that would leave it exactly as well off as its actual opportunity situation. Inequality is then evaluated under four states - each household’s own preferences and environment, and each replaced by a common reference - using the same measure, the same reference construction and the same numerical support in all four. The fully common state is the exhaustiveness test: it must be zero.

A two-player Shapley game: preferences against the complete non-preference environment.

$$\begin{array}{ll} I_{00} = 0.134 & I_{01} = 0.031 \\ I_{10} = 0.148 & I_{11} = 0.000 \end{array}$$

$$C_{\text{pref}} = \frac{1}{2}[(I_{00} - I_{10}) + (I_{01} - I_{11})] = 0.0085$$

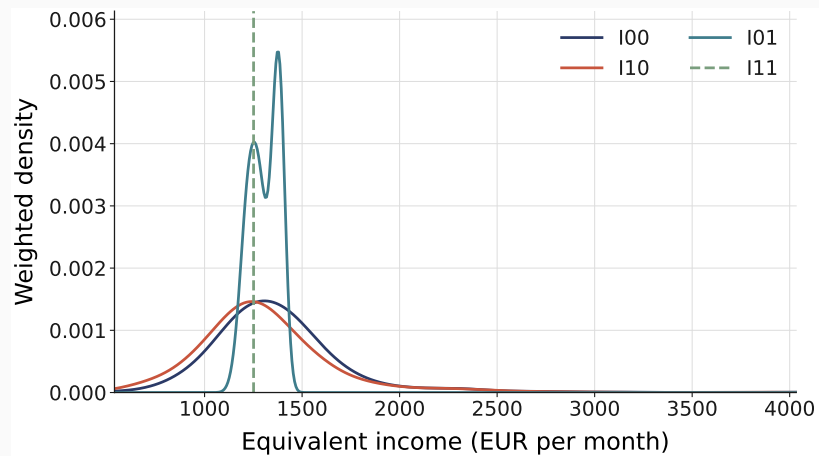
$$C_{\text{env}} = \frac{1}{2}[(I_{00} - I_{01}) + (I_{10} - I_{11})] = 0.126$$

equalizing the environment alone reduces inequality by 77%, equalizing preferences alone raises it by 10%; the Shapley attribution averages the two orders and gives the environment 94%.

the environment is then split by an Owen value into job access, earning opportunities, and household endowments and needs.

Equalizing preferences alone raises inequality slightly, from 0.134 to 0.148; equalizing the environment alone collapses it to 0.031; the fully common state is numerically zero. The Shapley value averages the two orders, which is what makes the attribution order-independent and exhaustive. Inside the environment, a grouped Owen value splits the contribution three ways. Two numbers, two questions: equalizing the environment by itself reduces inequality by 77 per cent, and equalizing preferences by itself raises it by 10 per cent; the Shapley value, which averages over the order in which preferences and environment are equalized, attributes 94 per cent to the environment. An attribution is never a quantity removed. Every share on this slide is on the raw welfare basis; the equalized basis is the companion, and the two are always named.

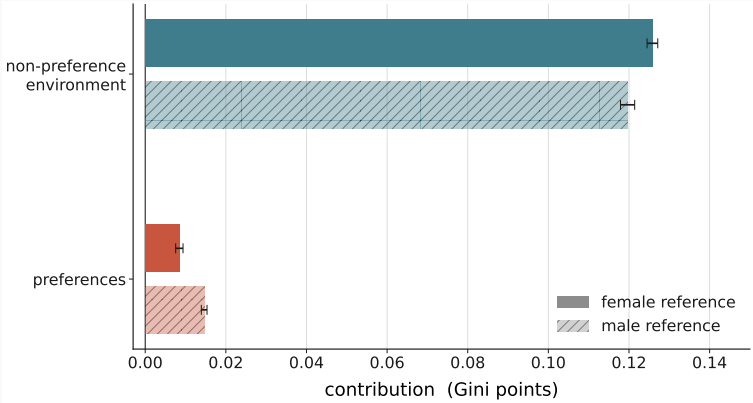
The distribution of equivalent income under the four states



the fully common state collapses to a point

Before the decomposition, the four states as distributions: own preferences and environment; preferences equalized, which widens the distribution; environment equalized, which nearly collapses it; and both, a point mass - the exhaustiveness test made visible.

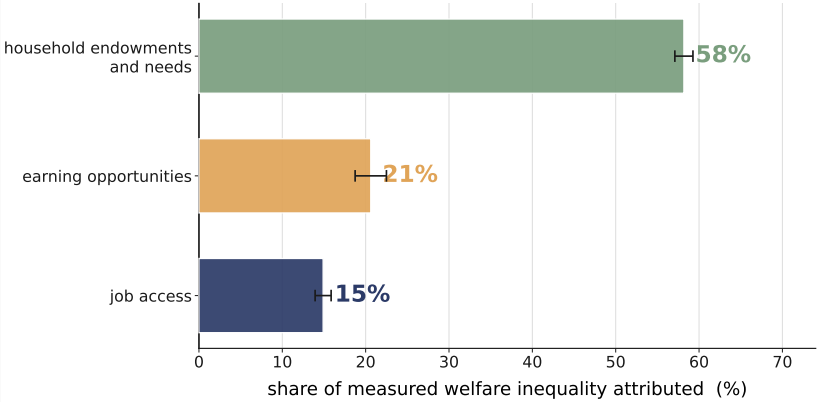
The non-preference environment accounts for 94% of measured welfare inequality; preferences for 6%.



93.7 ± 0.6 / 6.3 ± 0.6 female-primary; 89.1 / 10.9 male structural-zero; never averaged. Parameter intervals: environment [89.1, 95.8], preferences [4.2, 10.9].

Under the female-primary reference, the non-preference environment accounts for 93.7 per cent of measured money-metric inequality and preferences for 6.3; under the male structural-zero sensitivity, 89 and 11. The two conventions are reported as a pair and never averaged. The bands are integration precision; the parameter-uncertainty interval, propagated from the cluster-robust covariance, runs from 89 to 96 for the environment share and from 4 to 11 for preferences. The two kinds of uncertainty are never merged.

Household endowments and needs 58%, earning opportunities 21%, job access 15%; job opportunities together 36%.



integration bands ± 1.5 / ± 1.6 / ± 1.1 ; ordering identical under both reference conventions and under equalization.

Within the environment the budget side — non-labour income, household composition and their tax-benefit treatment — is the largest component at 58 per cent. Job access and earning opportunities together, the market-side contribution, account for 36 per cent of measured inequality. This ordering does not change across references or equivalence scales, and it is a structural, model-conditional result, not a claim that job opportunities are unimportant.

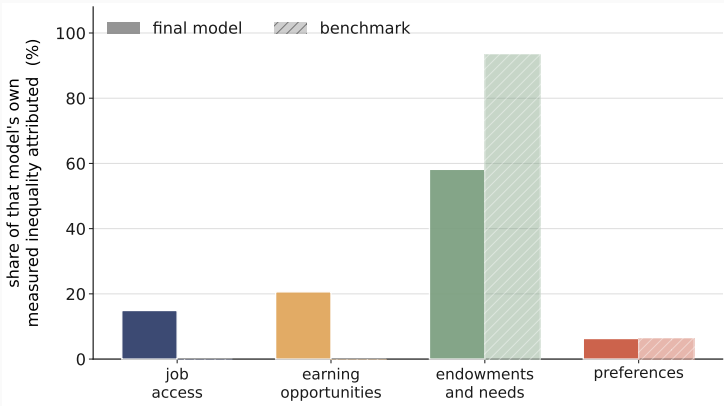
Unequal job opportunities account for about a third of measured welfare inequality; omitting them re-attributes almost nothing to preferences.

(1) job access + earning opportunities: $35.5\% \pm 1.0$ of measured inequality (the complete environment $93.7\% \pm 0.6$).

(2) Omitting heterogeneous opportunities: preference share $6.3\% \rightarrow 6.4\%$; measured inequality falls 24.2%; the market-side contribution is relabelled into endowments and needs.

To the first question: about a third of measured money-metric inequality is attributable to unequal job opportunities in the strict sense — access and earning opportunities — within an environment share of 94 per cent. To the second: a common-choice-set model does not turn opportunity into taste at the welfare level. Its preference share is 6.4 per cent against 6.3. What it does is understate measured inequality by a quarter and relabel the market-side contribution as endowments and needs. The absorption into tastes happens on the behavioural side, which is the next slide.

The benchmark fits the marginals as well, is 129 log-points worse, and recovers availability as taste.



35-hour coefficient: 2.58 as availability, 2.55 as preference; leisure-intercept sex gap $+0.43 \rightarrow -1.99$.

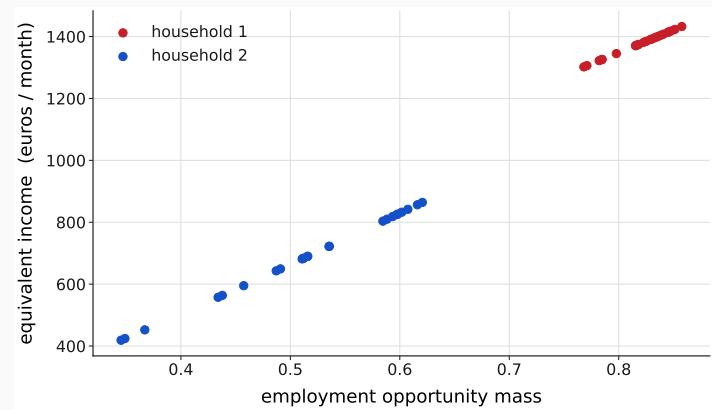
Estimated on the same data, the common-opportunity model matches the aggregate hours, employment and occupation shares as well as the RURO model does — in places better — while its likelihood is 129 nats worse. The availability constants come back as preference parameters, and the sex gap in the leisure intercept reverses sign. The welfare decomposition changes little in shares and much in level.

The preference share is the sensitive margin; the environment’s internal structure is stable.

perturbation →	preference share	environment ordering
reference convention	6.3 → 10.9	unchanged
age-curvature bounds	≈ +13%	unchanged
couples male-leisure pin	up to 73%	unchanged
50-400 drawn jobs	largest coefficient move 0.56 s.e. (0.2 s.e. from 100 to 400)	ordering unchanged

From 50 to 400 nested draws the largest coefficient move is half a standard error, and from 100 upward a fifth; no sign or ordering moves.

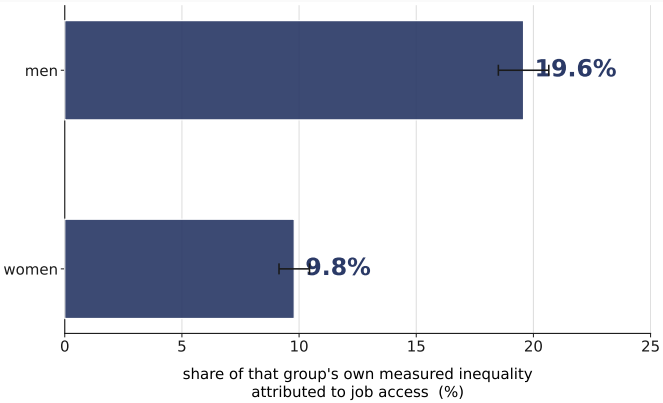
Within job access, nearly all of the household-varying contribution is geographic; job access matters twice as much for men.



Geography enters only through the employment margin, so a nested split attributes nearly all of the household-varying access contribution to the geographic block — but job access as a whole is 15 per cent of inequality. Holding an individual’s characteristics fixed and moving only the regional access environment moves money-metric welfare by up to a factor of two for a low-education profile. By sex, job access accounts for twice as large a share of men’s measured inequality as of women’s, a result that is stable across references.

geographic access 13% of inequality, job access 15%; men 20%, women 10%; structural, not causal.

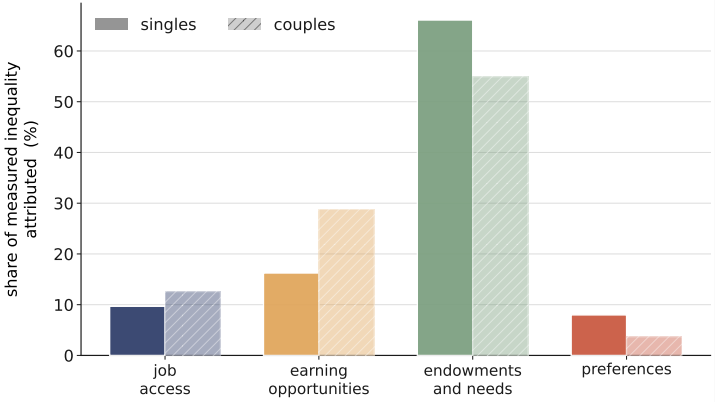
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Couples reproduce the same environment ordering; their preference share is not robustly identified.



2,275 couples; employment matched to four decimals; 35-hour peak 2.55 (women) and 3.29 (men). Cross-leisure interaction absent; welfare runs it at exactly zero — spouse leisure complementarity is not estimated here.

A clean both-flexible couples model converges and fits participation and hours by spouse closely. The environment decomposition has the same ordering as for singles. The couples preference share, however, moves by up to 73 per cent under admissible movement of the male leisure block, so singles remain the quantitative headline and couples the extension. One limitation to volunteer before it is asked: the cross-leisure interaction between the spouses' leisure is absent from this specification. It was not estimated and found small, and it was not estimated and then set to zero – it is not in the model, the welfare pipeline evaluates couples utility at exactly zero for it, and the omission is forced by identification. Read every couples number as conditional on that absence.

Each limit names the evidence that would lift it.

access vs capability	→	skill and credential data
persistent heterogeneity	→	panel choices or external moments
hours opportunities	→	desired-hours moments with a model counterpart
earning opportunities	→	administrative wage distributions
geography	→	local vacancy data

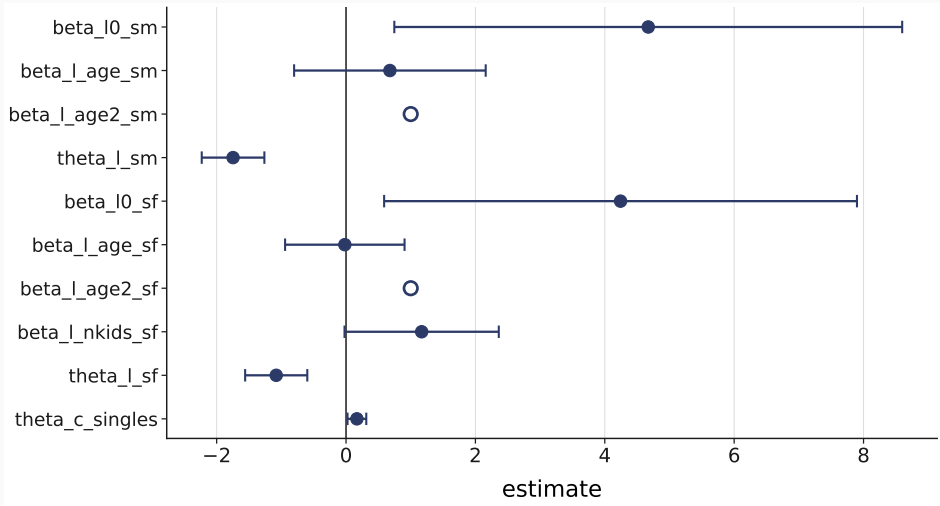
The paper’s limits are stated as scope. Each names the data that would identify what the current cross-section cannot: capability sets, persistent types, and a desired-hours counterpart. The lagged regional vacancy construction I tested added no detectable access signal at this resolution.

Method: a household-specific opportunity distribution estimated with preferences, priced through the real tax-benefit system.

Finding: the non-preference environment carries 94% of measured money-metric inequality; job opportunities about a third; the budget side more than the market side.

Margin: the preference share is the sensitive quantity, 6 to 11 per cent across normative conventions.

B1 — The estimated coefficients: preferences

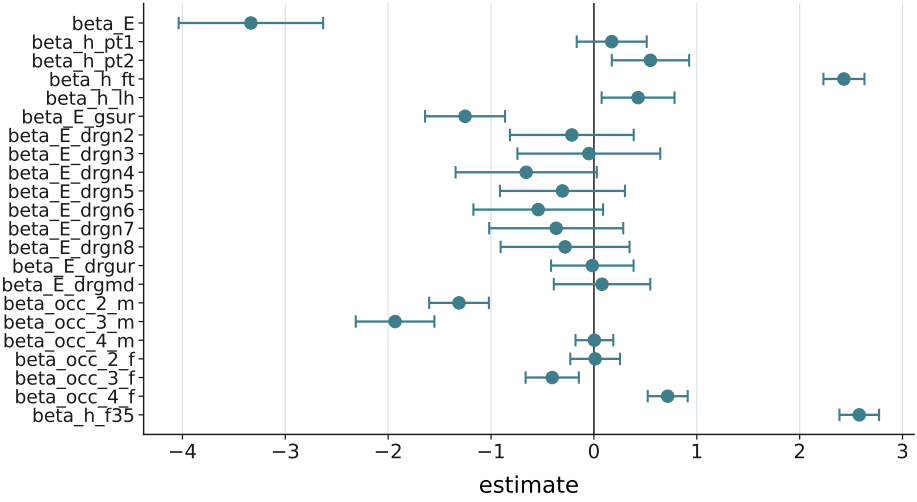


Held for: “What are the actual parameter estimates?”

The final singles model carries **41 estimated structural parameters**; this is the preference block. The full table with standard errors is in the paper’s Appendix A.

The leisure curvatures differ sharply by sex and are precisely estimated. Open markers rest on an active bound and carry no standard error; the ten pinned coordinates are not displayed.

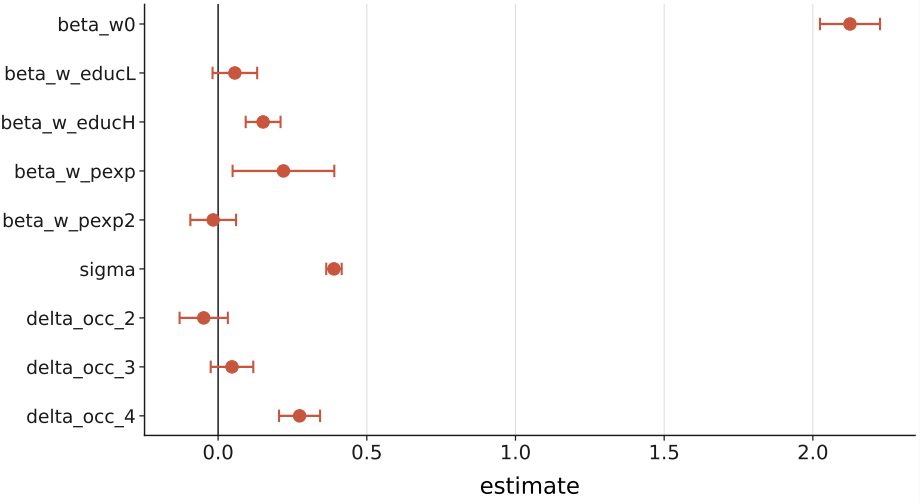
B1 — The estimated coefficients: job access



The job-access block: the group-specific unemployment rate, the region dummies and the urbanisation categories, plus the employment level.

This is the block whose dispersion the geographic split of slide 19 divides, and the reason almost all of the access channel's removable dispersion is geographic *by construction*: the remaining access block is conditioned on sex alone.

B1 — The estimated coefficients: earning opportunities

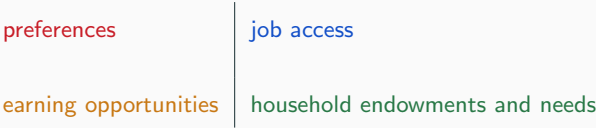


The earning-opportunity block: the occupation-availability weights and the wage-offer technology.

Read off the thing worth reading: the occupation-availability weights differ strongly by sex — men concentrated in and around routine manual, women's availability tilted toward nonroutine cognitive. That sex-specific structure is what keeps occupational sorting out of the taste block.

This is also the block the channel-relocation result of slide 4 perturbs: remove the occupation term here and the tertiary-education loading moves by six standard errors.

B2 — Exhaustiveness is a tested property



The fully common state is 0.000000 — numerically zero.

Held for: “*Why should I believe the four channels are exhaustive?*” — and it is the best question in the room.

This is a methodological result and it should be told as one, because the honest answer is that it did **not** close at first. An earlier implementation failed the exhaustiveness test on both limbs in every arm — equalising everything left an index **three times** the one it started from — and the headline was correctly **halted**.

Three things were found and removed. A missing budget channel, added as a fourth player formed from its own marginals rather than as a relabelled residual. An asymmetry in the money-metric inversion, where the reference side was frozen at the baseline while the attained side moved with the coalition; correcting it removed about 95 per cent of the residual. And a male reference block that was incomplete, because the specification carries a children-in-leisure term for women and none for men.

With all three fixed the fully common state is numerically zero. **The test was pre-stated before any cell was read.**

B3 — The scale moves with the channel that equalises needs

	raw	equivalized
preferences	6.30%	7.98%
opportunities and budgets	93.70%	92.02%
job access	14.90%	9.68%
earning opportunities	20.62%	16.24%
household endowments and needs	58.18%	66.10%
market side	35.51%	25.92%

Freezing the own scale leaves exactly the Gini of its inverse.

Held for: “You divide by an equivalence scale — doesn’t that break the decomposition?”

It does, under the natural convention, and exactly: freezing each household’s own scale in all states leaves a residual of 0.071799, which is **precisely the weighted Gini of one over the scale** — the same number to six decimals. Dividing a common numerator by a household-specific divisor reintroduces exactly the dispersion of the divisor.

The fix is the one the inversion already legislates: household composition is an endowments-and-needs object, so the scale moves with the channel that equalises composition, and exhaustiveness is restored exactly.

Equivalizing raises measured inequality by about a fifth and changes no qualitative conclusion, but it does move the market-side total from 35.51 to 25.92 per cent — both are reported and neither is “the” answer.

State the justification as **factor consistency, not closure**: the scale moves because household composition is itself part of the endowments-and-needs factor being equalised. **Exact closure is a consequence of this definition, not its justification.**

If instead the normative object were to equalise resources while leaving needs fixed, the household’s own scale would remain and the residual would be a genuine needs component requiring different accounting. That alternative is coherent; this paper adopts the first object.

ground	what it shows
definitional	the candidate's optimum tracks pay
identification	the hours block moves the moment by exactly 0
numerical	the level can agree; the profile cannot

Tried, and it fails on three independent grounds.

Held for: “*You have external data on who wants more hours — why not use it to discipline the hours block?*”

The honest answer is that it was tried and there is no counterpart to compare against. Three independent grounds.

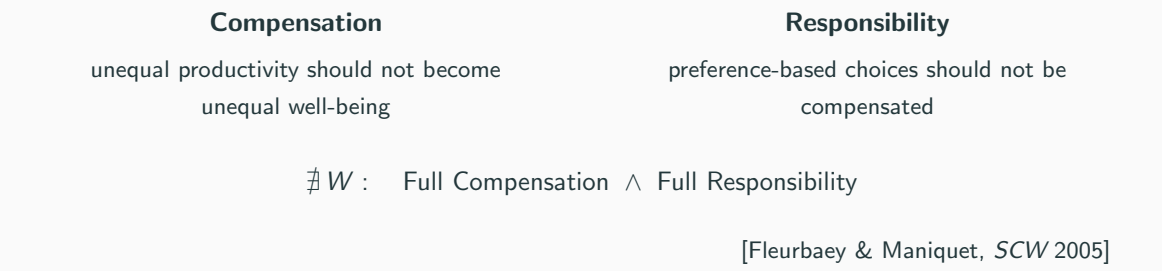
Definitional: the survey asks about more hours *at the current wage*, and the frame has no such menu — exactly one node per household carries the observed wage, the chosen one, so the candidate's optimum tracks pay, paying more for 98.1 per cent of workers at a median $2.58\times$ the observed wage.

Identification, and this is the decisive one: perturbing any of the six hours-opportunity coefficients moves the moment by **exactly zero** and moves not one household's optimum — removing the opportunity weights removes the very block the moment was meant to discipline.

Numerical: 24.3 per cent against 20.3, but the external share falls monotonically in hours and the model's does not, and the model puts none at long hours against 7.6 per cent.

The steelman was tried: restricting to the household's own ± 5 per cent wage neighbourhood fixes the pay artefact and brings the aggregate to 19.4 per cent — closer than the model deserves — and the hours block is *exactly as inert*.

The fork is architectural: a definition that removes the opportunity weights cannot identify the hours block, and one that retains them is the model's own choice distribution, not an unconstrained-desire object. So the external moment stays descriptive, and no calibration is proposed.



W^1 is one compromise — the one this paper takes to the data.

Haydar and Maniquet, *Jobs and Well-Being Measurement* (companion paper).

Held for: “Where does this welfare measure come from, and why this one?”

Reused from the companion theory talk; cited, not merged with the welfare slide.

Two principles we would like to respect. **Compensation:** people should not be penalized for unequal productive circumstances. **Responsibility:** people should be held responsible for their preferences, especially their tastes for work and leisure.

They are attractive separately but cannot both be imposed in full: no well-being measure satisfies Full Compensation and Full Responsibility together. So the literature is structured around compromises.

W^1 is one of them, and it is the one this paper takes to the data. Ignore the actual pay profile; put the same consumption level on all jobs in the individual’s ability set; among these feasible jobs the preference relation selects one; move the common level until being at that selected job is exactly indifferent to the actual bundle.

It satisfies Independence of $\mathbf{y}$ — it ignores the actual pay profile — and Responsibility for Equal Pay — once all feasible jobs pay the same, the choice among them reflects preferences over jobs.

The empirical translation replaces the ability set A with the *estimated opportunity density* g , which is where this paper’s work is.

	final model	benchmark
35-hour coefficient	2.58	2.55
	as availability	as preference
leisure-intercept sex gap	+0.43	-1.99
preference share	6.3%	6.4%

Job access and earnings are exactly zero here. By construction.

[2.5 min] This is the paper's own counterfactual about method and it is worth doing properly, because **the result is not the one the premise predicts** and saying so is what makes the rest credible.

Set it up in one line: every household-indexed argument of the availability object is replaced by a common value, so the benchmark is a conventional model with hours constants, a fixed cost of work and a sex-specific leisure block — what a careful analyst without an opportunity object would have fitted. On the benchmark side job access and earning opportunities are **exactly zero by construction**; those zeros are the benchmark's definition made arithmetic.

Then three findings, in this order. **First, fit does not separate them.** The benchmark is 129.1 nats worse on the objective (18151.85 against 18022.76 over 41 free parameters; a likelihood-ratio statistic of 258.2 on 25 degrees of freedom, quoted as an upper bound because the benchmark is not nested), but its marginal fit is *as good or better* — hours 0.0080 against 0.0083, occupation 0.0029 against 0.0034, wage quintiles 0.0203 against 0.0261. A referee judging on marginal fit alone would prefer the model with no opportunity object in it. That is the argument for not judging on marginal fit alone.

Second, the availability information goes into the taste parameters. The 5 hours-band constants reappear in the benchmark's *utility* index at essentially their availability values — mean absolute difference 0.037. The same constants, read twice. And the male-minus-female leisure gap **reverses sign**, from +0.428 to −1.991.

Third — the honest surprise — the preference share barely moves: 6.4 per cent under the benchmark against 6.3 under the preferred model. The omitted market-side share does not become taste. It mostly leaves the total — measured inequality falls -24.2 per cent raw and -15.3 equivalized — and what stays is relabelled onto endowments and needs.

Say the licensed sentence: *behavioural coefficients can absorb omitted availability structure; omitting heterogeneous opportunities materially lowers measured money-metric inequality; the welfare-decomposition preference share changes little; and the missing opportunity contribution is reallocated primarily within the non-preference environment rather than into preferences.* Do **not** claim it demonstrates a large welfare-level preference-misattribution effect.

25-MIN: compress to 1.5 — the reversed sex gap and the preference-share result.